

Solution Code - Module 3-Core Java - ThreadCreation

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q26_ThreadCreation.java

```
1 | // Method 1: Extending Thread class
2 | class ThreadA extends Thread {
3 |     @Override
4 |     public void run() {
5 |         for (int i = 1; i <= 5; i++) {
6 |             System.out.println("[Thread A (Extends Thread)] Message count: " + i);
7 |             try {
8 |                 Thread.sleep(100); // sleep for 100ms
9 |             } catch (InterruptedException e) {
10 |                 System.out.println("Thread A was interrupted.");
11 |             }
12 |         }
13 |     }
14 | }
15 |
16 | // Method 2: Implementing Runnable interface
17 | class RunnableB implements Runnable {
18 |     @Override
19 |     public void run() {
20 |         for (int i = 1; i <= 5; i++) {
21 |             System.out.println("[Thread B (Implements Runnable)] Message count: " + i);
22 |             try {
23 |                 Thread.sleep(120); // sleep for 120ms
24 |             } catch (InterruptedException e) {
25 |                 System.out.println("Thread B was interrupted.");
26 |             }
27 |         }
28 |     }
29 | }
30 |
31 | public class Q26_ThreadCreation {
32 |     public static void main(String[] args) {
33 |         System.out.println("=== Thread Creation Demo ===");
34 |
35 |         // Instantiating thread classes
36 |         ThreadA threadA = new ThreadA();
37 |         Thread threadB = new Thread(new RunnableB());
38 |
39 |         // Starting both threads (calls their run() methods in parallel)
40 |         System.out.println("Starting Thread A and Thread B...");
41 |         threadA.start();
42 |         threadB.start();
43 |     }
}
```

Solution Code - Module 3-Core Java - ThreadCreation

```
44 |          // Wait for both threads to finish to coordinate main output
45 |          try {
46 |              threadA.join();
47 |              threadB.join();
48 |          } catch (InterruptedException e) {
49 |              System.out.println("Main thread was interrupted.");
50 |          }
51 |
52 |          System.out.println("Both threads have finished execution.");
53 |      }
54 | }
```